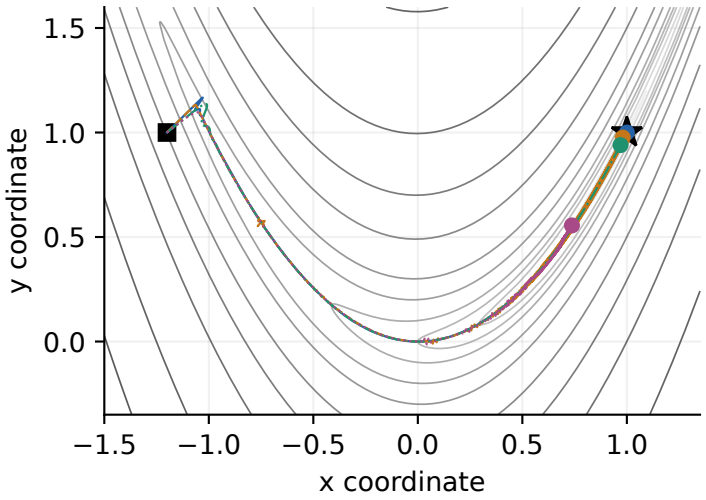
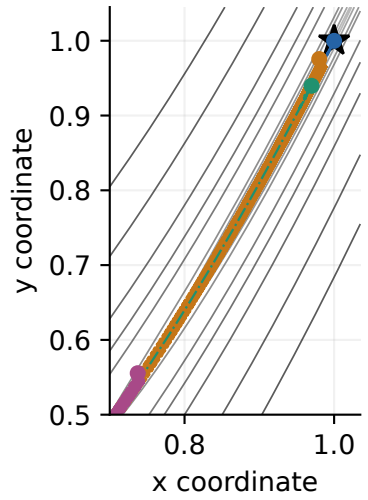
Controlled Rosenbrock: momentum × decoupled decay

Run028: Momentum, no decay
Run029: No momentum, no decay
Run030: Momentum, decay 0.1
Run031: No momentum, decay 0.1

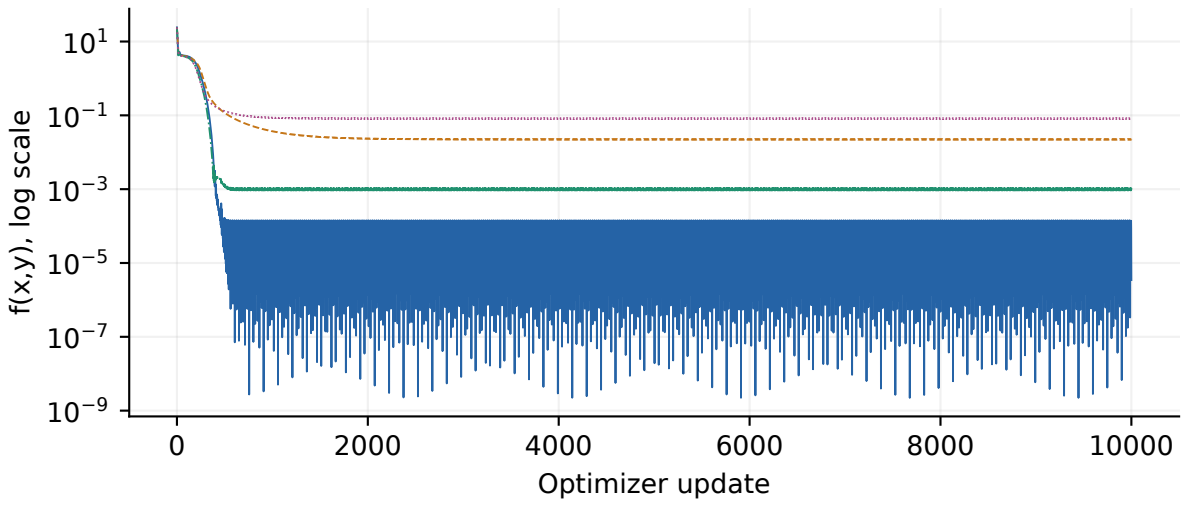
(a) Full trajectories: shared start



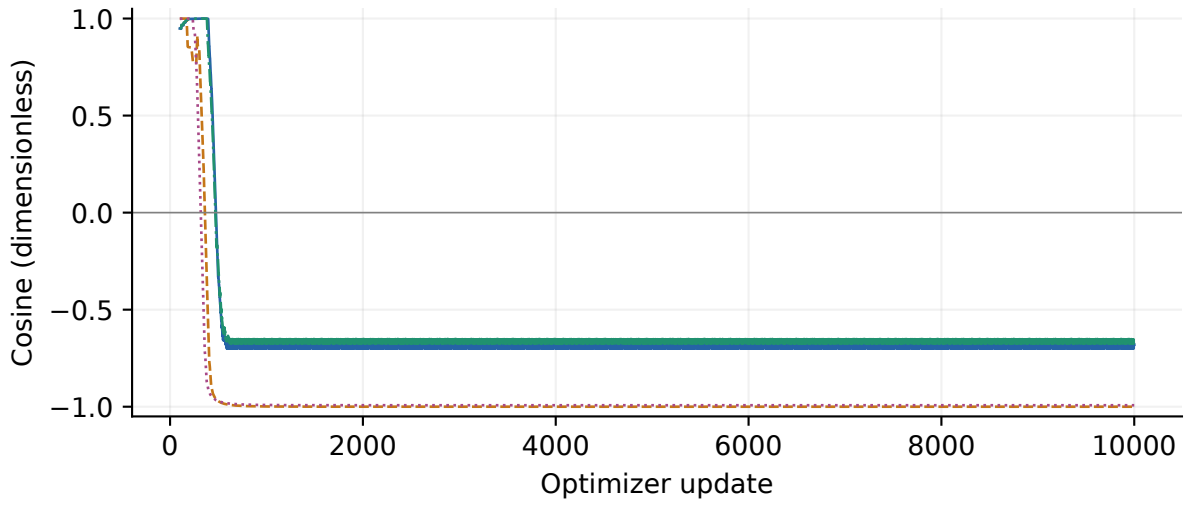
(b) Close-up near the minimum: final circles



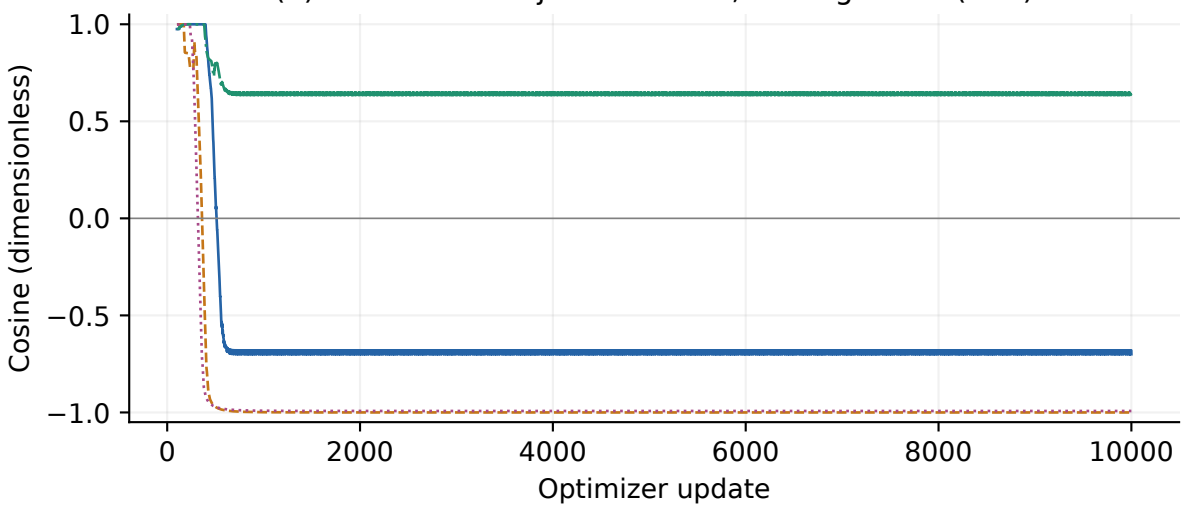
(c) Objective after update



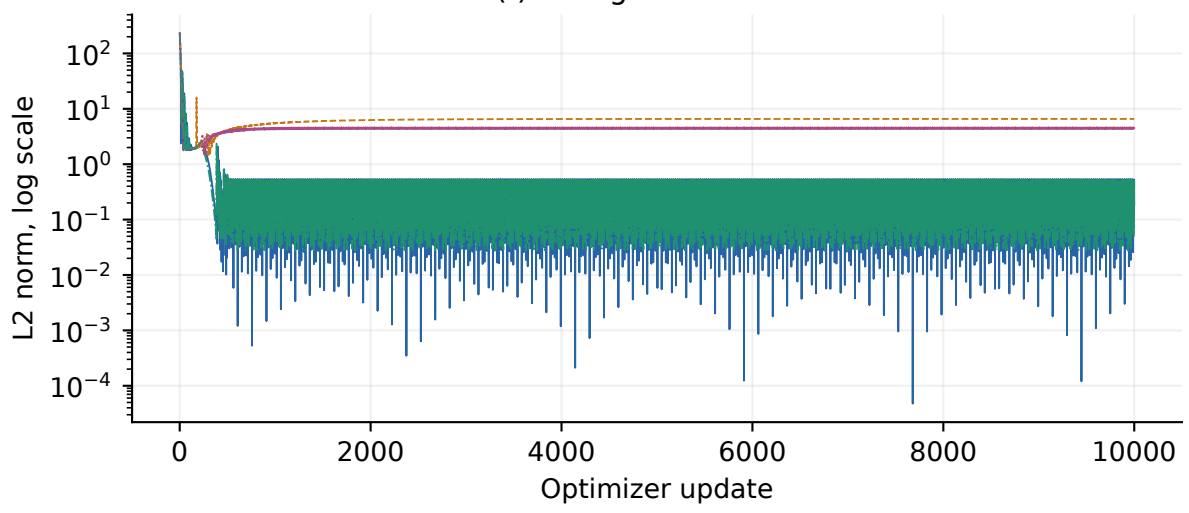
(d) Raw adjacent cosine, trailing mean (100)



(e) Momentum adjacent cosine, trailing mean (100)



(f) Raw gradient norm



All 10,000 updates and identical (-1.2,1) starts; LR=0.01, beta2=0.95, epsilon=1e-8.
Independent trajectories; one objective/start/LR. Final loss is not proof of sustained convergence. Raw cosine alone does not diagnose loss improvement.